Chapter 1 - Rings and Ideals

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Exercise 1

Suppose $x^r = 0$. Notice

$$\begin{aligned}(1+x)\left(1-x+x^2-x^3+\ldots+(-1)^{r-1}x^{r-1}\right) &= 1-x^r \\ &= 1\end{aligned}$$

So $1+x$ is a unit in A. $\square$

Exercise 2

(i) (a) ($\Rightarrow$): Let $g = b_0 + b_1x + \ldots + b_mx^m$ be the inverse of f. Now

$$(a_0 + a_1x + \ldots + a_nx^n)(b_0 + b_1x + \ldots + b_mx^m) = 1$$

By checking the coefficients of $x^{n+m}, x^{n+m-1}, x^{n+m-2}\ldots$, we get

$$\begin{cases} a_nb_m = 0 \\ a_n^2b_{m-1} = 0 \\ a_n^3b_{m-2} = 0 \\ \vdots \\ a_n^mb_1 = 0 \\ a_n^{m+1}b_0 = 0 \end{cases}$$

On the other hand, by checking the constant term directly,

$$a_0b_0 = 1$$

Therefore a_0 is unit, and $a_n^{m+1} = a_n^{m+1}b_0a_0 = 0$. So a_n is nilpotent.
Moreover, $-a_nx^n$ is a nilpotent element in $A[x]$. By *Ex.1*,

$$f - a_nx^n = a_0 + a_1x + \ldots + a_{n-1}x^{n-1}$$

is a unit in $A[x]$, so by iterating the process above we'll see $a_{n-1}, \ldots, a_1$ are nilpotent.

(b) ($\Leftarrow$): By *Ex.1*, the sum of a unit and a nilpotent element is a unit. Notice $a_1x, a_2x^2, \ldots, a_ns^n$ are nilpotent in $A[x]$ while a_0 is a unit in $A[x]$, so their sum f is a unit in $A[x]$.

$\square$

(ii) (a) ($\Rightarrow$) : Consider the constant term of power of f, from which we know a_0 is nilpotnet. Nilpotent elements are closed under addition, so

$$a_1x + a_2x^2 + \ldots + a_nx^n = f - a_0$$

is nilpotent, i.e.

$$a_1 + a_2x + \ldots + a_nx^{n-1}$$

is nilpotent. By induction we can have $a_0, a_1, \ldots, a_n$ being nilpotent.

(b) $(\Leftarrow)$: When $a_0, a_1, \dots, a_n$ are all nilpotent, let $k \in \mathbb{Z}^+$ be sufficiently large, and obviously we can finally reach $f^k = 0$.

$\square$

(iii) (a) $(\Rightarrow)$: Suppose we have $g = b_0 + b_1 x + \dots + b_m x^m$ such that $fg = 0$.
If $b_m = 0$, we're done. Otherwise, similar to *(i)*, by checking the coefficients of $x^{m+n}, x^{m+n-1}, \dots$ respectively, we'll obtain

$$\begin{cases} a_n b_m = 0 \\ a_{n-1} b_m = 0 \\ a_{n-2} b_m^2 = 0 \\ \vdots \\ a_1 b_m^{n-1} = 0 \\ a_0 b_m^n = 0 \end{cases}$$

So $f \cdot b_m^n = 0$.

(b) $(\Leftarrow)$: Trivial.

(iv) (a) $(\Leftarrow)$: Let $g = b_0 + b_1 x + .. + b_m x^m$. Since g is primitive, $\exists \lambda_0, \lambda_1, \dots, \lambda_m \in A, s.t.$

$$\lambda_0 b_0 + \lambda_1 b_1 + \dots + \lambda_m b_m = 1$$

Therefore

$$\lambda_0 b_0 f + \lambda_1 b_1 f + \dots + \lambda_m b_m f = f$$

Also, suppose

$$\mu_0 a_0 + \mu_1 a_1 + \dots + \mu_n a_n = 1$$

Since

$$\begin{aligned} fg = \lambda_0\mu_0 b_0 a_0 &+ \lambda_1\mu_0 b_1 a_0 x + \lambda_2\mu_0 b_2 a_0 x^2 + \dots + \lambda_m\mu_n b_m a_n x^{m+n} \\ &+ \lambda_0\mu_1 b_0 a_1 x + \lambda_1\mu_1 b_1 a_1 x^2 + \dots \\ &+ \lambda_0\mu_2 b_0 a_2 x^2 + \dots \\ &\vdots \end{aligned}$$

So

$$(\lambda_0\mu_0 b_0 a_0) + (\lambda_0\mu_1 b_0 a_1 + \lambda_1\mu_0 b_1 a_0) + (\lambda_0\mu_2 b_0 a_2 + \lambda_1\mu_1 b_1 a_1 + \lambda_2\mu_0 b_2 a_0) + \dots + (\lambda_m\mu_n b_m a_n) = 1$$

(b) $(\Rightarrow)$: When fg is primitive. We prove by contradiction and suppose g is not primitive. Since fg is primitive, we may assume

$$b_0 a_0 \xi_0 + (b_1 a_0 + b_0 a_1)\xi_1 + (b_2 a_0 + b_1 a_1 + b_0 a_2)\xi_2 + \dots + b_m a_n \xi_{m+n} = 1$$

Therefore

$$\begin{aligned} &(a_0\xi_0 + a_1\xi_1 + \dots + a_n\xi_n) b_0 \\ &+ (a_0\xi_1 + a_1\xi_2 + \dots + a_n\xi_{n+1}) b_1 \\ &+ (a_0\xi_2 + a_1\xi_3 + \dots + a_n\xi_{n+2}) b_2 \\ &\vdots \\ &+ (a_0\xi_m + a_1\xi_{m+1} + \dots + a_n + \xi_{m+n}) b_m = 1 \end{aligned}$$

Contradictory.

$\square$

Remark 1. *Consider the sum of terms in each of $n+m-1$ anti-diagonals*

$$\begin{array}{cc} & \begin{array}{ccccc} \lambda_0 & \lambda_1 & \lambda_2 & \cdots & \lambda_m \end{array} \\ \begin{array}{c} \mu_0 \\ \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{array} & \begin{bmatrix} b_0a_0 & b_1a_0x & b_2a_0x^2 & \cdots & b_ma_0x^m \\ b_0a_1x & b_1a_1x^2 & b_2a_1x^3 & \cdots & b_ma_1x^{m+1} \\ b_0a_2x^2 & b_1a_2x^3 & b_2a_2x^4 & \cdots & b_ma_2x^{m+2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_0a_nx^n & b_1a_nx^{n+1} & b_2a_nx^{n+2} & \cdots & b_ma_nx^{m+n} \end{bmatrix} \end{array}$$

Exercise 3

We can prove it by induction. View $A[x_1, \ldots, x_n]$ as a ring over $A[x_1, \ldots, x_{n-1}]$ in an indeterminate x_n. $\square$

Exercise 4

Recall that the Jacobson radical is the intersection of all maximal ideals, while the nilradical is the intersection of all prime ideals. We'll prove that every prime ideal $p \subseteq A[x]$ is maximal:

$$\forall \text{prime ideal } p, \text{ it's maximal}$$

$\Leftrightarrow$

$$\forall \text{ prime ideal } p, \text{ Krull dimension of } p \text{ is } 0$$

$\Leftrightarrow$

$$\dim A[x] = 0$$

$\Leftrightarrow$(By *[Algebraic Geometry - Hartshorne]Proposition 1.7*)

$$\dim \{\emptyset\} = 0, \ \{\emptyset\} \text{ in affine 1-space } A$$

Which is trivial.

Remark 2. *By induction, we'll get Jacobson radical being equal to nilradical in $A[x_1, x_2, \ldots, x_n], \forall n \in \mathbb{Z}^+$*

Exercise 5

(i) (a) $(\Rightarrow)$: Suppose $fg = 1$, where

$$g = \sum_{n=0}^{\infty} b_n x^n$$

By checking the constant term of fg, we get

$$a_0 b_0 = 1$$

Thus a_0 is a unit in A.

(b) $(\Leftarrow)$: Notice

$$(a_0 + a_1x + \ldots)(a_0^{-1} - a_1a_0^{-2}x + (a_1^2a_0^{-3} - a_2a_0^{-2}) + \ldots) = 1$$

Such construction of f^{-1} can goes forever, so f is a unit in $A[[x]]$.

(ii) (a) When f is nilpotent, obviously a_0 is nilpotent. Nilpotent elements closed under addition, so $f - a_0$ is nilpotnet, indicating a_1 being nilpotent. Iterate the process above and we get $a_0, a_1, a_2, \ldots$ being nilpotent.

(b) No. Counterexample can be reached on *MathStackExchange.*

(iii) (a) $(\Leftarrow)$: For a_0 in maximal ideal I, we can easily check that

$$\mathfrak{I}_I = \left\{ b_0 + b_1x + b_2x^2 + \ldots \mid b_0 \in I, b_1, b_2, \ldots \in A \right\}$$

Is a maximal ideal in $A[[x]]$. Therefore $f = \sum_{n=0}^{\infty} a_n x^n \in \mathfrak{I}_I$.

So when a_0 is in the Jacobson Radical of A, f must be in the Jacobson Radical of $A[[x]]$.

(b) $(\Rightarrow)$: We only need to prove the maximal ideals of $A[[x]]$ are all in the form

$$\mathfrak{I}_I = \left\{b_0 + b_1x + b_2x^2 + \ldots \mid b_0 \in I, b_1, b_2, \ldots \in A\right\}$$

where I is a maximal ideal of A.

Indeed, it's easy to see that the constant terms of all the elements in a arbitary ideal $\mathfrak{I}$ of $A[[x]]$ forms an ideal in A, we denote then by I and let $\mathfrak{I}_I = \mathfrak{I}$. When $\mathfrak{I}_I$ is maximal, I must be maximal in A, otherwise $\mathfrak{I}_I$ can expand to a larger ideal $\mathfrak{I}_{\overline{I}}$ where $\overline{I}$ is a maximal ideal containing I.

Hence the ideal of $A[[x]]$ is maximal if and only if the constant term ideal is maximal in A. And f is in a arbitary maximal ideal will indicate a_0 belongs to the correspond maximal ideal. Deduction of Jacobson Radical follows immediately.

(iv) Proved in *(iii)* above.

(v) **(UNFINISHED)** *(I'm not sure whether I understood this problem correctly.)*

Consider

$$\begin{aligned} \varphi : A &\longrightarrow A[[x]] \\ a &\longrightarrow a + 0x + 0x^2 + \ldots \end{aligned}$$

And define φ^{-1} by taking the constant term in the formal power series. For prime ideal $\mathfrak{p} \subseteq A$,

$$\mathfrak{p}A[[x]] = \left\{a_0 + a_1x + a_2x^2 + \ldots \mid a_0, a_1, \ldots \in \mathfrak{p}\right\}$$

We can prove it's a

Remark 3. *The proof in **Ex.1.5(ii)** is identical to the one in **Ex.1.2(ii)***

Exercise 6

(UNFINISHED)

By *Proposition 1.11*, if the nilradical is equal to the Jacobson radical, since any prime ideal would contain the Jacobson radical, i.e. intersection of all maximal ideals, it would contain some of those maixmal ideals.

Therefore the problem is equivalent to prove that every prime ideals in A is maximal,

Exercise 7

For any prime ideal $\mathfrak{p} \subseteq A$, consider the map

$$A \longrightarrow A/\mathfrak{p}$$

From the problem we know $\forall x \in A, \exists n > 0, \text{s.t. } x^n = x$. So $x(x^{n-1} - 1) = 0$, and we obtain

$$\overline{x}(\overline{x}^{n-1} - \overline{1}) = 0$$

We may assume $x \notin \mathfrak{p}$, i.e. $\overline{x} \neq 0$. Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is a integral domain. So $\overline{x}^{n-1} = \overline{1}$, $\overline{x}$ has its inverse in $A/\mathfrak{p}$. Because x was an arbitary element in $A\backslash\mathfrak{p}$, $A/\mathfrak{p}$ is a field. Hence $\mathfrak{p}$ is maximal. $\square$

Exercise 8

Consider the intersection of all prime ideals of A, which is the nilradical and obviously minimal with respect to inclusion. $\square$

Exercise 9

Recall that $\text{Rad}(\mathfrak{a})$ is the intersection of all prime ideals containing $\mathfrak{a}$. The proof can be obtained directly from it. $\square$

Exercise 10

(a) $(i) \Rightarrow (ii)$: If A has exactly one prime ideal $\mathfrak{p}$, such ideal will be the unique maximal ideal of A.

Recall that, by *Corollary 1.5*, every non-unit of A lies in $\mathfrak{p}$, therefore nilpotent. On the other hand, others will be the uni of A.

(b) $(ii) \Rightarrow (iii)$: $\forall$non-zero $\overline{a} \in \overline{A}, \exists a^{-1} \in A\backslash\mathfrak{R}$, s.t. $aa^{-1} = 1$. And we have $\overline{a}\overline{a^{-1}} = \overline{1}$.

(c) $(iii) \Rightarrow (i)$: $A/\mathfrak{R}$ is a field if and only if $\mathfrak{R}$ is a maximal ideal. Because a nilradical is the intersection of all prime ideals, we can see $\mathfrak{p}$ is unique and maximal in A.

$\square$

Exercise 11

(i) $2x = 4x^2 = 4x$, so $2x = 0$.

(ii) For any prime ideal $\mathfrak{p}$, consider the map

$$\begin{aligned}\varphi : A &\longrightarrow A/\mathfrak{p}\\ x &\longrightarrow \overline{x}\end{aligned}$$

Since $x^2 = x$, $\overline{x}^2 = \overline{x}, \overline{x}(\overline{x} - 1) = 0$. $A/\mathfrak{p}$ is a integral domain, so $\overline{x} = 0$ or $\overline{x} = 1$. Hence $A/\mathfrak{p}$ only contains two elements, i.e. $\overline{1}$ and $\overline{0}$, and it's a field for sure.

(iii) For $I = (x, y)$, consider $z = x + y + xy$, we'll prove $(z) = (x, y)$.
By definition, $(z) \subseteq (x, y)$. Also, notice

$$zx = (x + y + xy)x = x + xy + xy = x$$

So $x \in (z)$. In the same way we have $y \in (z)$, thus $(z) = (x, y)$. Rest of the proof can be done by induction.

$\square$

Exercise 12

Suppse we have a nontrivial idempotent e. By $e^2 = e$ we have

$$e(1 - e) = 0$$

So both e and $1-e$ is a zero-divisor, hence they're not unit and lies in the unique maximal ideal. Therefore $1 = e + (1 - e)$ is in the maximal ideal, contradictory.

Exercise 13

(POOR-WRITTEN UNCLEAR PROOF)

We prove by contradiction: Suppose $\mathfrak{a} = (1)$, then $\exists$polynomial $F \in A[f(x_f)]$, for all f and x_f, such that

$$F(f(x_f), \ldots) = 1, \text{ finitely many } f \in \Sigma$$

For those F, we pick out one with the least number of $f \in \Sigma$ being involved.

For a specific $f_0 \in \Sigma$ which was involved in the polynomial F above, consider the terms of f_0 with its highest degree α_0 in $F(f(x_f), \ldots)$. Because $F(f(x_f), \ldots) = 1$, those terms with degree α_0 sums up to 0, i.e.

$$f(x_f)^{\alpha_0}(\text{rest of the terms}) = 0$$

Apparently $f(x_f)^{\alpha_0} \neq 0$, so the rest of the terms sums up to 0. Then for the rest of the terms, we can again similarly pick out a f_1, α_1 and sort out "rest of the terms" to be 0. We iterate the process above. Since there're only finitely many terms, the process must stop, making a contradiction.

(We don't need to worry about characteristic of K is finite or not, since w.l.o.g. we can let those terms in F which adds up over the characteristic vanish in the first place.)

Exercise 14

(i) By Zorn's lemma, Σ has maximal elements.

(ii) $\forall$maximal $I \subseteq \Sigma$, suppose $a, b \in A$, s.t. $ab \in I$. I consists of zero-divisors, so $\exists u \in I$, s.t. $(ab)u = 0$. Then

$$a(bu) = 0$$

indicating a being a zero-divisor, so does b. Since I is maximal, $a \in I$ (or $b \in I$)

(iii) This can be obtained immediately by taking the union of every maximal element in Σ

Exercise 15

Trivial. *(See [Abstract Algebra - D&F(3rd Edition)]Chap 12, [Algebraic Geometry - Hartshorne]Sect 1)*

Exercise 16

(i) $\operatorname{Spec}(\mathbb{Z}) = \{(p) \mid p \in \mathbb{Z} \text{ is prime }\}$

(ii) $\operatorname{Spec}(\mathbb{R}) = \{(0)\}$

(iii) $\operatorname{Spec}(\mathbb{C}[x]) = \{(p(x)) \mid p(x) \in \mathbb{C}[x], \deg p = 1\}$

(iv) $\operatorname{Spec}(\mathbb{R}[x]) = \{(p(x)) \mid p(x) \in \mathbb{R}[x] \text{ prime (or we can say irreducible) }\}$

(v) **(UNFINISHED)**
$\operatorname{Spec}(\mathbb{Z}[x]) = \{\}$

Exercise 17

(UNFINISHED) Obviously they form a basis of open sets for Zariski topology.

(i) Trivial.

(ii) Trivial.

(iii) Trivial.

(iv) Trivial.

(v) Without loss of generation ww may only conside X convered by basic open sets X_{f_i}, since every general open sets can be in the form of union of basis.

Now

$$\forall \bigcup_{k \in I} X_{f_k} \supseteq X = \operatorname{Spec}(A) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcup_{i=1}^{n} X_{f_{k_i}} \supseteq X$$

$\Leftrightarrow$

$$\forall \bigcap_{k \in I} V(f_k) \subseteq \emptyset \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_{k_i}) \subseteq \emptyset$$

$\Leftrightarrow$

$$\forall \bigcap_{k \in I} V(f_k) = \emptyset \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_{k_i}) = \emptyset$$

$\Leftrightarrow$

$$\forall (f_1, f_2, \ldots) = (1) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } (f_{k_1}, \ldots, f_{k_n}) = (1)$$

When $\forall (f_1, f_2, \ldots) = (1)$, $\exists g_{k_1}, \ldots, g_{k_n} \in A, \text{s.t. } 1 = \sum a_{k_i} f_{k_i}, a \in A$, which implies $(1) = (f_{k_1}, \ldots, f_{k_n})$.

(vi) Similarly,

$$\forall \bigcup_{k \in I} X_{f_k} \supseteq X_f \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcup_{i=1}^{n} X_{f_{k_i}} \supseteq X_f$$

$\Leftrightarrow$

$$\forall \bigcap_{k \in I} V(f_k) \subseteq V(f) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_{k_i}) \subseteq V(f)$$

$\Leftrightarrow$

$$\forall f_1, f_2, \ldots \text{s.t. } f \in (f_1, f_2, \ldots) \implies \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } f = a_1 f_1 + \ldots + a_n f_n, (a_k \in A)$$

Which is trivial.

(vii) Since $\{X_f\}$ forms a basis for $\mathrm{Spec}(A)$, $\exists \text{famliy} \mathcal{F}, \text{s.t. } X_0 = \bigcup_{f\in\mathcal{F}} X_f$. Now

$$X_0 \text{ quasi compact}$$

$\Leftrightarrow$

$$\forall X_{f_1}, X_{f_2}, \ldots \in A, \text{s.t. } \bigcup_{k=1}^{\infty} X_{f_k} \supseteq \bigcup_{f\in\mathcal{F}} X_f, \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcup_{i=1}^{n} X_{f_{k_i}} \supseteq \bigcup_{f\in\mathcal{F}} X_f$$

$\Leftrightarrow$

$$\forall f_1, f_2, \ldots, \text{s.t. } \bigcap_{k=1}^{\infty} V(f_k) \subseteq \bigcap_{f\in\mathcal{F}} V(f), \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \bigcap_{i=1}^{n} V(f_k) \subseteq \bigcap_{f\in\mathcal{F}} V(f)$$

$\Leftrightarrow$

$$\forall f_1, f_2, \ldots \in A, \text{s.t. } \mathcal{F} \subseteq (f_1, f_2, \ldots), \exists f_{k_1}, \ldots, f_{k_n}, \text{s.t. } \mathcal{F} \subseteq (f_{k_1}, \ldots, f_{k_n})$$

(i) $(\Rightarrow)$: We prove by contradiction and suppose $\mathcal{F}$ can only be infinite, then let $f_1, f_2 \ldots$ be the elements in $\mathcal{F}$, and such $f_{k_1}, \ldots, f_{k_n}$ cannot exist.

(ii) $(\Leftarrow)$: When $\mathcal{F}$ is finite, it is a direct proposition from *(vi)*.

$\square$

Exercise 18

(i) $\{x\}$ is closed if and only if the only prime ideal containing $\mathfrak{p}_x$ is $\mathfrak{p}_x$ itself, which is equivalent to $\mathfrak{p}_x$ being maximal.

(ii) Trivial.

(iii) $y \in \{\overline{x}\} \Leftrightarrow \mathfrak{p}_y \in V(\mathfrak{p}_x) \Leftrightarrow \mathfrak{p}_x \subseteq \mathfrak{p}_y$.

(iv) *(What's the definition of "neighborhood" here? I'll interpret it as "closed neighborhood" anyway.)*

Consider $V(x), V(y)$, which is a neighborhood of x, y respectively. Soppose $x \in V(\mathfrak{p}_y), y \in V(\mathfrak{p}_x)$, then by *(iii)* we get $\mathfrak{p}_x = \mathfrak{p}_y$, contradictory.

$\square$

Exercise 19

(Beautiful!)

In ring A, we consider the category $\mathscr{C}$ of ideals, morphisms umder inclusions. We take its subcategory $\mathscr{C}_0$ of prime ideals, i.e.$\mathrm{Obj}(\mathscr{C}_0) = \mathrm{Spec}(A)$.

In this case, the initial object of $\mathscr{C}$ is the nilradial of A, and it lies in $\mathscr{C}_0$ if and only if the nilradical is prime. Therefore, $\mathscr{C}_0$ has a initial object, i.e. the partially ordered set of $\mathrm{Spec}(A)$ consists of a connected tree instead of a union of disjoint trees, if and only if $\mathrm{Nil}(A)$ is prime.

On the other hand, for any two subsets of A, denoted by E, F, we denote the complement of $V(E), V(F)$ by X_E, X_F respectively. We can have

$$X_E \cap X_F \neq \emptyset$$

$\Leftrightarrow$

$$V(E) \cup V(F) \neq \mathrm{Spec}(A)$$

$\Leftrightarrow$

$$\mathrm{Nil}(A) \subseteq \mathrm{Spec}(A)$$

(Becuase $V(\mathrm{Nil}(A)) = \mathrm{Spec}(A) = X$. Otherwise when nilradical is not prime, $\mathscr{C}_0$ consists of distinct trees with each root $r_1, r_2, \ldots$, from which we can let $E = \{r_1\}, F = \{r_2, \ldots\}$ and have $V(E) \cup V(F) = X$.)

Exercise 20

(SKIPPED) *(I haven't learnt topology yet.)*

Exercise 21

(UNFINISHED)

$$\begin{tikzcd} A \arrow[r, "\phi"] \arrow[d, "V()"] & B \arrow[d, "V()"] \\ X = \mathrm{Spec}(A) & Y = \mathrm{Spec}(B) \arrow[l, "\phi^*"'] \end{tikzcd}$$

(i) For $\mathfrak{q} \in Y_{\phi(f)}$,

$$\mathfrak{q} \in \phi^{*-1}(X_f)$$

$\Leftrightarrow$

$$\phi^*(\mathfrak{q}) \in X_f$$

$\Leftrightarrow$

$$\phi^*\mathfrak{q} \notin V(f)$$

$\Leftrightarrow$

$$f \notin \phi^*(\mathfrak{q}) = \phi^{-1}(\mathfrak{q})$$

$\Leftrightarrow$

$$\phi(f) \notin \mathfrak{q}$$

$\Leftrightarrow$

$$\mathfrak{q} \in Y_{\phi_{(f)}}$$

So $\phi^{*-1}(X_f) = Y_{\phi(f)}$.

(ii) For any prime ideal $\mathfrak{q} \in Y$,

$$\mathfrak{q} \in \phi^{*-1}(V(\mathfrak{a}))$$

$\Leftrightarrow$

$$\phi^*(\mathfrak{q}) \in V(\mathfrak{a})$$

$\Leftrightarrow$

$$\mathfrak{a} \subseteq \phi^*(\mathfrak{q})$$

$\Leftrightarrow$

$$\phi(\mathfrak{a}) \subseteq \phi(\phi^*(\mathfrak{q})) = \mathfrak{q}$$

$\Leftrightarrow$

$$\mathfrak{q} \in V(\phi(\mathfrak{a})) = V(\mathfrak{a}^e)$$

So $\phi^{*-1}(V(\mathfrak{a})) = V(\mathfrak{a}^e)$.

(iii) $\forall \mathfrak{q} \in V(\mathfrak{b})$

$$\phi^{-1}(\mathfrak{b}) \subseteq \phi^*(\mathfrak{q})$$

$\Leftrightarrow$

$$\mathfrak{b} \subseteq \mathfrak{q}$$

Which is true by definition, so $\overline{\phi^*(V(\mathfrak{b}))} \subseteq V(\phi^{-1}(\mathfrak{b}))$. **(UNFINISHED)**

(iv) (a) From *(i)* we already know ϕ^* is continuous. Since we now have ϕ being surjection, ϕ^* is obviously a bijection. Also, the continuity of derives directly from *(iii)*.

(b) In particular, let $Y = A/\mathfrak{R}$ and ϕ be the canonical homomorphism.

(v) We directly prove the "More precisely" part:

$$\phi^*(Y) \text{ dense in } X$$

$\Leftrightarrow$

$$\overline{\phi^*(Y)} = X \tag{1}$$

$\Leftrightarrow$

$$\overline{\phi^*(V(\mathrm{Nil}(B)))} = X$$

$\Leftrightarrow$

$$\overline{\phi^*(V(\mathrm{Nil}(B)))} = V(\mathrm{Nil}(A))$$

$\Leftrightarrow$ (by *(iii)*)

$$V(\mathrm{Nil}(B)^c) = V(\mathrm{Nil}(A))$$

$\Leftrightarrow$

$$V(\phi^{-1}(\mathrm{Nil}(B))) = V(\mathrm{Nil}(A))$$

$\Leftrightarrow$

$$\mathrm{Rad}(\phi^{-1}(\mathrm{Nil}(B))) = \mathrm{Nil}(A) \tag{2}$$

To prove ((1) $\Leftrightarrow$ (2)), we only need

$$\mathrm{Ker}(\phi) \subseteq \mathrm{Rad}(\phi^{-1}(\mathrm{Nil}(B)))$$

$\Leftarrow$

$$\mathrm{Ker}(\phi) \subseteq \phi^{-1}(\mathrm{Nil}(B))$$

$\Leftrightarrow$

$$0_B \subseteq \mathrm{Nil}(B)$$

Which is trivial.

(vi) Trivial.

(vii)

Exercise 22

(a) $\forall k \in [1, n] \cap Z$,

$$X_k = \{A_1 \times \ldots \times A_{k-1} \times \mathfrak{p} \times A_{k+1} \times \ldots \times A_n \mid \mathfrak{p} \in \mathrm{Spec}(A_k)\}$$

So we can define a map

$$\varphi : \mathrm{Spec}(A_k) \longrightarrow X_k$$

by

$$\mathfrak{p} \longmapsto A_1 \times \ldots \times A_{k-1} \times \mathfrak{p} \times A_{k+1} \times \ldots \times A_n$$

and the homeomorphic property is trivial in the Zariski topology defined in this way.

(b) (I) $(ii) \Rightarrow (i)$: From *(a)* we see

$$X_1 = \{\mathfrak{p} \times A_2 \mid \mathfrak{p} \in \mathrm{Spec}(A_1)\}, X_2 = \{A_1 \times \mathfrak{p} \mid \mathfrak{p} \in \mathrm{Spec}(A_2)\}$$

So $X_1 \cap X_2 = \emptyset$.

(II) $(iii) \Rightarrow (ii)$: Suppose we have $e \in A$, s.t. $e^2 = 2, e \notin \{0, 1\}$. Then $A \cong eA \times (1-e)A$.

(III) $(i) \Rightarrow (iii)$:

$$X \text{ disconnect}$$

$\Leftrightarrow$

$$X \text{ is a union of disjoint nontrivial clopen set}$$

$\Leftrightarrow$

$$\exists \text{ideal } I, J, \text{s.t. } V(I) \cap V(J) = V(I+J) = \emptyset, V(I) \cup V(J) = V(IJ) = V(I \cap J) = X$$

When such I, J exist, let $a \in I, b \in J$, s.t. $a + b = 1$. Since $V(IJ) = X, IJ \subseteq \mathrm{Nil}(A)$, so $\exists n \in \mathbb{Z}^+$, s.t. $(ab)^n = 0$. Consider

$$s_1 = \mathrm{C}_{2n}^0 a^{2n} + \mathrm{C}_{2n}^1 a^{2n-1} b + \ldots + \mathrm{C}_{2n}^{n-1} a^{n+1} b^{n-1}$$

$$s_2 = \mathrm{C}_{2n}^{2n} b^{2n} + \mathrm{C}_{2n}^{2n-1} a b^{2n-1} + \ldots + \mathrm{C}_{2n}^{n+1} a^{n-1} b^{n+1}$$

Now

$$s_1 + s_2 = s_1 + s_2 + \mathrm{C}_{2n}^n a^n b^n = (a+b)^{2n} = 1$$

and

$$s_1 s_2 = 0$$

So

$$s_1(1 - s_1) = 0$$

s_1, s_2 are nontrivial idempotent.

$\square$

Exercise 23

(i) X_f is open by definition, and we only have to prove that X_f is closed.
For any nonzero, non-unital $f \in A$ consider $1 - f$. Notice:

$$\begin{aligned} V(f) \cap V(1-f) &= V((f) + (1-f)) \\ &= V(1) \\ &= \emptyset \end{aligned}$$

$$\begin{aligned} V(f) \cup V(1-f) &= V((f)(1-f)) \\ &= V(0) \\ &= X \end{aligned}$$

Therefore $V(f) = X_{1-f}$.

(ii) Because

$$\bigcap_{k=1}^{n} V(f_k) = V\left(\sum_{k=1}^{n} (f_k)\right)$$

where the ideal $\sum_{k=1}^{n}(f_k)$ is finitely generated, therefore principal by *Ex-1.11(iii)*. So $\exists f \in A$, s.t.

$$(f) = (f_1, f_2, \ldots, f_n) = \sum_{k=1}^{n} (f_k)$$

and therefore

$$V(f) = \bigcap_{k=1}^{n} V(f_k)$$

(iii) For any clopen subset $Y \subseteq X$, let $Z = X \backslash Y$. By definition Z is also clopen. From *Prop-1.11(i)*,

$$\forall \mathfrak{p} \in Y, \mathfrak{p} \backslash \left(\bigcup_{\mathfrak{q} \in Y \backslash \mathfrak{p}} \mathfrak{q} \right)$$

is nonempty. Thus we can pick out $f \in \mathfrak{p} \backslash \left(\bigcup_{\mathfrak{q} \in Y \backslash \mathfrak{p}} \mathfrak{q} \right)$ and consider $(1 - f)$: From *(i)* we know $V(f), V(1-f)$ is a partition of X, so by $f \in \mathfrak{p}, \mathfrak{p} \in Y$ we have

$$V(f) = Y, V(1-f) = Z$$

But since $f \notin \bigcup_{\mathfrak{q} \in Y} \mathfrak{q}$, $V(f) \subsetneq Y$, contradictory.

(iv) X is compact by *Ex-17(iv)*.
To prove X is Hausdorff, we need to prove for $\forall \mathfrak{p}, \mathfrak{q} \in X, \mathfrak{p} \neq \mathfrak{q}$,

$$\exists \text{open set } X_1, X_2, \text{s.t. } \mathfrak{p} \in X_1, \mathfrak{q} \in X_2, X_1 \cap X_2 = \emptyset$$

$\Leftrightarrow$

$$\exists \text{closed set } V_1, V_2, \text{s.t. } \mathfrak{p} \notin V_1, \mathfrak{q} \notin V_2, V_1 \cup V_2 = X$$

$\Leftrightarrow$

$$\exists \text{closed set } V_1, V_2, \text{s.t. } \mathfrak{p} \in V_2 \backslash V_1, \mathfrak{q} \in V_1 \backslash V_2, V_1 \cup V_2 = X$$

Such V_1, V_2 can be obtained by

$$V_1 = X \backslash \{\mathfrak{q}\}, V_2 = X \backslash \{\mathfrak{p}\}$$

$\square$

Exercise 24

(a) (I) For addition:

(i) $+$ is commutative because $\wedge, \vee$ are both commutative.

(ii)

$$\begin{aligned} a + 0 &= (a \wedge 1) \vee (a' \wedge 0) \\ &= a \vee 0 \\ &= a \end{aligned}$$

(iii)

$$\begin{aligned} a + a' &= (a \wedge a') \vee (a' \wedge a) \\ &= 0 \vee 0 \\ &= 0 \end{aligned}$$

(iv)

$$\begin{aligned} (a + b) + c &= (ab' \vee a'b) + c \\ &= (ab' \vee a'b)c \vee (ab + a'b')c \\ &= abc \vee a'bc \vee ab'c \vee abc' \end{aligned}$$

Which is symmetric with respect to a, b, c, therefore addition is associative.

(II) For multiplication:

(i)

$$ab = a \wedge b = b \wedge a = ba$$

(ii)

$$a \cdot 1 = a \wedge 1 = a$$

(iii) Associative of multiplication is given by the associativity of $\vee$.

(iv)

$$\begin{aligned} a(b + c) &= a(bc' \vee b'c) \\ &= abc' \vee ab'c \end{aligned}$$

$$\begin{aligned} ab + ac &= ((a' \vee b')ac) \vee (ab(a' \vee c')) \\ &= ab'c \vee abc' \end{aligned}$$

And we obtian the distributive law.

(III) For Boolean property: $a^2 = a \wedge a = a$.

(b) We'll prove one side, the other side cna be obtained by duality.

(i) Associativity:

$$\begin{aligned} a \vee (b \vee c) &= a \vee (b + c + bc) \\ &= a + b + c + bc + ab + ca + abc \end{aligned}$$

$$\begin{aligned} (a \vee b) \vee c &= (a + b + ab) \vee c \\ &= a + b + c + ab + ca + bc + abc \end{aligned}$$

So $(a \vee b) \vee c = a \vee (b \vee c)$.

(ii) Commutavity: Because multiplication is commutative.

(iii) Absorption:

$$\begin{aligned} a \vee (a \wedge b) &= a \vee (ab) \\ &= a + ab + aab \\ &= 3a \\ &= a \end{aligned}$$

(iv) Identity: $a \vee 0 = a$

(v) Distributivity:

$$\begin{aligned} a \vee (b \wedge c) &= a \vee (bc) \\ &= a + bc + abc \end{aligned}$$

$$\begin{aligned} (a \vee b) \wedge (a \vee c) &= (a + b + ab)(a + c + ac) \\ &= a + ac + ac + ab + bc + abc + ab + abc + abc \\ &= a + bc + abc \end{aligned}$$

So $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$.

(vi) Complements:

$$\begin{aligned} a \vee a' &= a + (1 - a) + a(1 - a) \\ &= 1 \end{aligned}$$

$\square$

Exercise 25

(Beautiful!)

Every Boolean lattice, namely L, is isomorphic to $Y = \{X_f \mid f \in A(L), X = \mathrm{Spec}(A(L))\}$

Exercise 26